Deterministic Choice Probabilities and Share Inversion for Factor Probit with Many Alternatives

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Abstract

Multinomial probit choice probabilities are high-dimensional Gaussian integrals with no closed form, and the standard tool — the GHK simulator — is noisy, per-alternative, and effectively unusable for large choice sets; the practical consequence is that flexible substitution is routinely surrendered for logit tractability. We present a deterministic transform for the factor-structured case $\Sigma = VV^\top + D$: conditionally on k Gaussian factors the alternatives are independent, so a lattice survival-product identity prices *all* N alternatives in $O(NL)$ per quadrature node, with no simulation anywhere. On benchmark problems the transform's error is flat at $\sim 3 \times 10^{-4}$ from $N = 5$ to $N = 5000$ (22 seconds at $N = 5000$) while GHK's error grows and its cost crosses over by $N \approx 200$; at matched accuracy GHK would need on the order of twelve hours at $N = 5000$. Because the map is smooth and deterministic, derivatives are exact rather than traded between noise and bias, and two capabilities follow that simulation-based practice does not offer: market-share inversion (the probit analogue of the BLP contraction; $N = 1000$ in under a minute, matching targets four orders of magnitude below their own Monte Carlo noise) and the full single-removal assortment ensemble from one pass. We map the method's honest boundary with a spectral-decay sweep over dense covariances: the factor floor decays slowest at intermediate spectra, yet at the tested size the rank-8 transform matches or beats GHK at $R = 10^4$ throughout — so the boundary is conditional (accuracy demands finer than the affordable factor floor), not territorial. On a ground truth misspecified for every candidate model, factor structure carries the first half of substitution fidelity and matching the idiosyncratic noise family a further factor of two.

1 Introduction

The multinomial probit (MNP) model prices choice among N alternatives with jointly normal utilities, and thereby allows arbitrary substitution patterns — the property that logit-family models purchase tractability by giving up. Its probabilities, however, are $(N-1)$-dimensional Gaussian orthant integrals. The field's workhorse for three decades has been the GHK simulator [1, 2, 3]: importance sampling along a Cholesky factor, one alternative at a time. GHK is a fine estimator of a single probability at moderate N, but it has three structural liabilities. Its error decays as $R^{-1/2}$ in the number of draws; its cost grows with N both per-alternative and in the number of alternatives to price; and its output is a simulation, so derivative-based estimation

*Every number in this paper is produced by a committed, seeded script with correctness anchors run before any comparison: `https://github.com/microprediction/kinetics` (experiments 13 and 14). The algorithm ships in the `thurstone` package (`https://github.com/microprediction/thurstone`); package–benchmark parity is itself a reported number.

must either accept noisy gradients or freeze draws (common random numbers) and accept the frozen noise as bias. The practical result is visible across empirical work: probit is avoided at scale, and its substitution flexibility with it.

This paper presents a deterministic alternative for the factor-structured case

$$U_i \;=\; \mu_i + v_i^\top f + \sqrt{D_i}\,\varepsilon_i, \qquad f \sim N(0, I_k), \quad \varepsilon_i \sim N(0,1) \text{ i.i.d.}, \tag{1}$$

i.e. $\Sigma = VV^\top + \mathrm{diag}(D)$ — the form in which applied covariance structure usually arrives (market factors, nest-like loadings, random coefficients on a few characteristics). The method composes two old ideas. First, conditionally on the factors the alternatives are independent, so the choice probability is a one-dimensional integral — the classical reduction. Second, on a lattice the whole field of N such integrals collapses to one pass: the field survival product is built once and each alternative's leave-one-out field is recovered by division [6]. A k-dimensional Gauss–Hermite (or scrambled-Sobol) rule around that $O(NL)$ inner loop gives every choice probability, deterministically, with cost linear in N per node.

Three consequences matter more than the forward speed itself. *Smoothness*: the map is analytic in μ, so gradients are exact — the estimation-relevant property GHK cannot have. *Invertibility*: market-share inversion, the engine of demand estimation since [7] and essentially never attempted for probit at scale, becomes a coordinate-Newton fixed point that converges in a handful of forward passes. *Ensembles*: the same conditional field pass yields every single-removal (assortment, stockout, delisting) counterfactual at once.

We benchmark against a validated GHK implementation with correctness anchors run before any comparison (Section 3), and we map the honest boundary of the method (Section 4) with a spectral-decay sweep over dense covariances — where we expected a clear GHK-wins regime and report what we actually found. Related work is discussed in Section 5; estimation implications, including implicit differentiation through the inversion, in Section 6.

2 The transform

Work in min-wins convention ($X_i = -U_i$; the winner has the smallest performance). Let the base density of ε_i have CDF Φ and density ϕ (results extend to arbitrary, alternative-specific base densities; only the factors need be Gaussian). Conditional on f, alternative i's performance has location $m_i(f) = -\mu_i + v_i^\top f$ (sign flipped by convention) and scale $\sqrt{D_i}$, and the alternatives are independent, so

$$p_i \;=\; \mathbb{E}_f\left[\int \frac{1}{\sqrt{D_i}}\,\phi\Big(\tfrac{x-m_i(f)}{\sqrt{D_i}}\Big)\prod_{j\neq i} S_j(x \mid f)\,dx\right], \qquad S_j(x \mid f) = 1 - \Phi\Big(\tfrac{x-m_j(f)}{\sqrt{D_j}}\Big). \tag{2}$$

On a lattice of L points the product over $j \neq i$ for *all* i costs one pass: form $\log S_{\text{field}} = \sum_j \log S_j$ once and recover each leave-one-out field by subtraction. The factor expectation uses product Gauss–Hermite nodes for $k \leq 4$ and scrambled-Sobol quasi–Monte Carlo beyond (deterministic given the seed; both smooth in μ). Total cost $O(Q\,N\,L)$ for all N probabilities, with Q the node count ($\sim$145 at $k = 2$).

Inversion. Given target shares $\hat{p}$ (and V, D), find μ with $p(\mu) = \hat{p}$. Holding the field frozen, each alternative's probability is a monotone function of its own location, so a damped coordinate-

Newton step

$$\mu_i \leftarrow \mu_i - \frac{\log p_i(\mu) - \log \hat{p}_i}{\partial \log p_i / \partial \mu_i}$$

with the field refrozen each iteration converges in a handful of passes; the own-location slope $\partial p_i/\partial\mu_i = \mathbb{E}_f \int z\phi(z)/D_i \cdot (\text{rest field})\, dx$ comes from the same lattice pass as p. Two details matter in practice: warm-start from the independent-field inverse (using each alternative's total variance), and a tail-aware tolerance — alternatives with shares below $O(1/N)\cdot 10^{-4}$ are matched best-effort, since their locations are unidentified in the same sense that never-winners' abilities are. The design is the original transform's inversion [6] generalized per node; the warm start and the observation that the choice-space Jacobian is intrinsically well-conditioned regardless of $\mathrm{cond}(\Sigma)$ come from portfolio calibration practice with the same race [11].

Removal ensembles. Deleting alternative i is a marginal operation: survivors' conditional fields lose one factor of S_i, recovered by division. One conditional field pass therefore prices $P(j \text{ wins} \mid i \text{ removed})$ for all pairs (i, j).

3 Benchmark against GHK

All numbers from committed seeded scripts; anchors first. On the binary case, where MNP has the closed form $\Phi(\Delta\mu/\sigma_\Delta)$, the lattice transform agrees to 3×10^{-16} and our GHK implementation — validated before being compared against — to 2×10^{-14} at $R = 2 \times 10^5$. At $N = 5$ both methods sit within the noise of a 10^7-draw Monte Carlo reference. The shipped `thurstone.FactorRace` agrees with the benchmark kernel to 1.1×10^{-5}, so the claims attach to the library.

N	lattice transform	GHK ($R = 1000$)
5	21 ms, err 4.1×10^{-4}	1 ms, err 3.1×10^{-3}
20	95 ms, err 3.8×10^{-4}	12 ms, err 4.4×10^{-3}
50	257 ms, err 2.1×10^{-4}	77 ms, err 7.7×10^{-3}
200	930 ms, err 2.8×10^{-4}	1333 ms, err 6.7×10^{-3}
1000	4.5 s, err 8.0×10^{-4} *	—
5000	**22.6 s, err** 9.0×10^{-4} *	—

Table 1: Full share vector, $k = 2$ factors, versus 2×10^6-draw MC truth (*at the reference's own noise floor, $\sim 4 \times 10^{-4}$). The lattice error is flat in N; GHK's grows, and its cost crosses over by $N \approx 200$.

Matched accuracy. GHK's error scales as $R^{-1/2}$. Extrapolating its cost by the *measured* power law over the tested range ($\alpha \approx 2.0$, a lower bound — the asymptotic per-share cost is cubic), GHK at $N = 5000$ and $R = 1000$ needs $\geq$13 minutes for $\sim 7 \times 10^{-3}$ accuracy; matching the lattice's 9×10^{-4} requires $R \approx 55{,}000$, i.e. **on the order of twelve hours against 22 seconds**, under assumptions favorable to GHK.

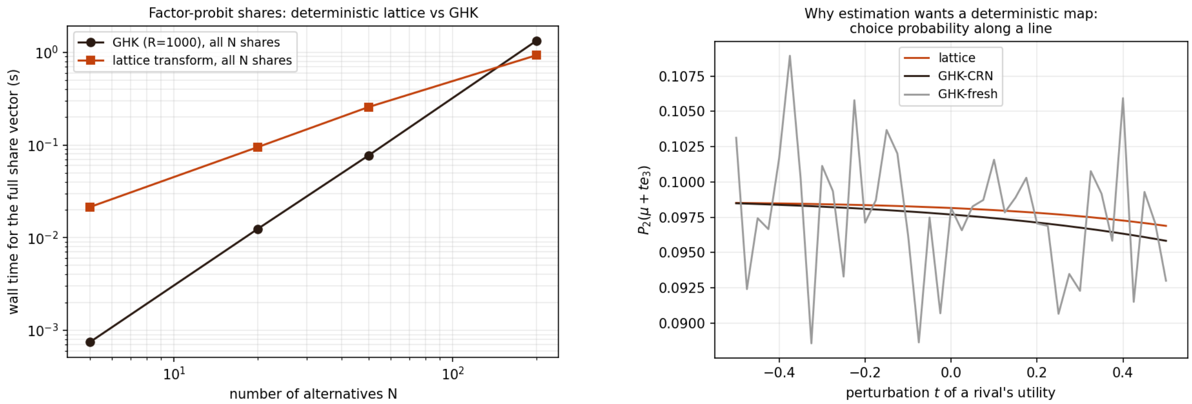

Figure 1: Left: wall time for the full share vector. Right: a choice probability along a line in utility space. Fresh-draw GHK is unusable for derivatives (curvature noise 39.3 vs. 0.01); common random numbers make GHK smooth *by freezing the noise in as bias* — the CRN curve visibly sits off the truth — while the lattice is smooth and unbiased. GHK trades noise for bias; the lattice needs no trade.

Share inversion. At $N = 1000$, $k = 2$, with target shares from 5×10^6 MC draws (no inverse crime): the coordinate-Newton inversion completes in **59 seconds**, matching the targets to 6.3×10^{-9} — four orders of magnitude below the targets' own sampling noise — and recovering utilities of identified alternatives to 0.014. To our knowledge no simulation-based practice attempts probit share inversion at this scale.

Assortment ensemble. All 200 single-removal share vectors from one conditional pass, 3.7× faster than recomputation and identical to 5.6×10^{-17}.

4 The honest boundary

Full covariance. The transform requires $\Sigma \approx VV^\top + D$; GHK does not. Sweeping random correlation matrices with eigenvalue decay $\lambda_m \propto m^{-\gamma}$ at $N = 50$ (Figure 2, left): the factor floor decays fastest for near-low-rank truth ($\gamma = 3$: 5.6×10^{-2} at $k = 1$ down to 1.2×10^{-3} at $k = 8$) and slowest at *intermediate* decay ($\gamma = 1.5$: 2.4×10^{-2} down to 2.6×10^{-3}) — the same mid-spectrum hardness seen for kinked kernels in the physics companion experiments. We expected a clear GHK-wins regime and, at this size and accuracy, did not find one: at every tested decay the $k = 8$ lattice matches or beats GHK at $R = 10^4$ (errors 3.2–4.1×10^{-3}) while being deterministic and cheaper. The honest boundary is therefore conditional rather than territorial: GHK's arbitrary-Σ generality wins where required accuracy is finer than the affordable factor floor (intermediate spectra with k capped), and at very small N where its raw cost is lowest.

Substitution fidelity. Why want probit-style flexibility at all? On a ground truth misspecified for *every* candidate — $t(5)$-distributed factors, skew-normal idiosyncratic noise — we calibrate plain logit, factor mixed logit, and factor probit to identical full-menu shares (factor loadings supplied) and score deletion counterfactuals against fresh simulation truths (Figure 2, right): deletion blocks whose removed share sits at the Monte Carlo noise floor are uninformative (all models tie, as they must); on the informative blocks the ordering is clean and consistent. Removing the 42.5%-share favorite: plain logit misallocates 20.8% of the redistributed mass, factor mixed logit 14.6%, factor probit **7.0%**. Over mid-mass blocks (2–10% removed): 30.8%, 25.4%, 13.6% respectively. Factor structure carries the first half of the correction; the idiosyncratic law carries a further factor of two here — with the honest caveat that the truth's skew-normal idiosyncratic noise is closer to the Gaussian family, so this margin should be read as "matching the noise family matters," not "probit always wins."

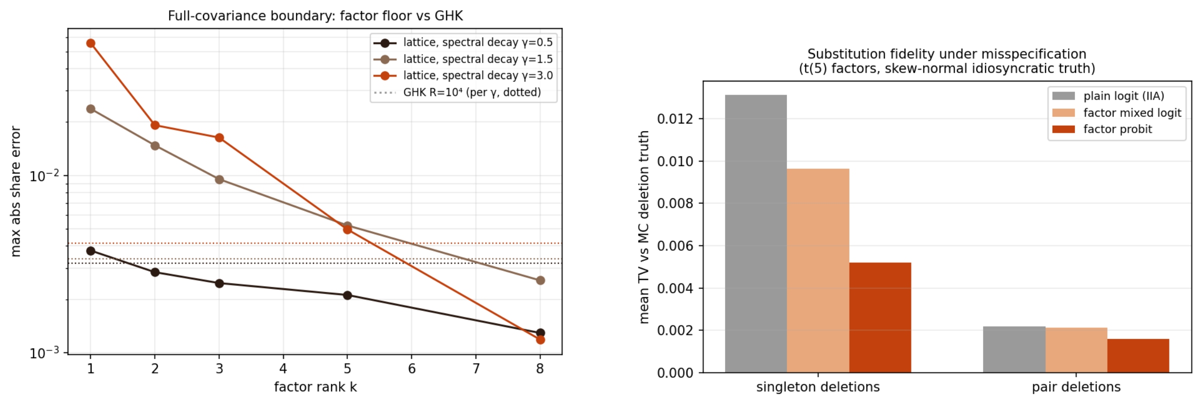

Figure 2: Left: the factor floor versus GHK under spectral decay γ of the true covariance. Right: substitution fidelity under misspecification.

5 Related work

GHK [1, 2] and its use in maximum simulated likelihood are standard [3]; analytic approximation lines include MACML [4]. The one-dimensional reduction for *independent* probit is classical, and dimensional reduction under factor structure is noted in [5]; the contribution here is the assembled algorithm — the $O(NL)$ all-alternatives inner loop via the field-product/divide-out identity [6], deterministic factor quadrature around it, the coordinate- Newton share inversion with its conditioning and warm-start structure [11], and the one-pass removal ensemble — together with the benchmark and boundary evidence. Share inversion and its acceleration are central to the BLP literature [7, 8]; recent work makes logit-based BLP automatically differentiable [9], and a 2026 preprint attacks MNP probabilities with trained neural amortization [10] — evidence of current demand; the present method is deterministic, training-free, and invertible. Mixed logit as the tractable route to flexible substitution is textbook [3]; Section 4 quantifies when the noise law matters relative to factor structure.

6 Discussion

Three directions follow. First, estimation: because the forward map is smooth and deterministic, the share inversion admits implicit differentiation — gradients of an outer likelihood or GMM objective with respect to structural parameters pass through the fixed point via

$d\mu/d\theta = -(\partial p/\partial\mu)^{-1}\,\partial p/\partial\theta$ — giving, in effect, automatically differentiable probit demand estimation; the logit analogue is recent [9], the probit version is open. Second, the same machinery prices races with non-Gaussian and alternative-specific idiosyncratic noise (only the factors need be Gaussian), including a Gumbel base that reproduces Luce exactly at zero loadings and yields correlated-softmax generalizations. Third, scale: the inner loop is a bandwidth-bound tensor contraction, well suited to accelerators; and the inversion inherits the original transform's interpolation structure (per-node location-to-probability curves are cross-correlations, computable at all offsets at once), which we have not yet exploited.

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